

SmartCare v0.1 - Initial Engineering Brief and AI Activity Card

A2 Case Study Stage 1 student resource

SmartCare scenario

SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The client says: 'We need software to help manage patients, practitioners and appointments.' This is not yet a complete specification.

Initial Engineering Brief

1. Problem summary

Write approximately 100 words.

- SmartCare Community Clinic uses spreadsheets and paper records for managing patients and appointments. its a time consuming process and maintaining it is also difficult and may lead to errors such as missing or incorrect information. It also needs a software system improvisation for the management of patients, practitioners, and appointments details. The system should also allow staff to store and access relevant information efficiently and make the appointment system easier. Additional requirements need to be gathered to understand exactly what information, functions, rules, security, and user access the system should provide.

2. Initial stakeholders

Stakeholder	Possible need
Patients	Easy booking, cancellation
practitioners	Access patient info, history, appointment timew
Admin staff	Managing booking, easy navigation of the system
IT support	Maintaining the system efficiently

3. Initial features

Feature	Confirmed or provisional?	Why?
Booking appointment	confirmed	Main function of the system or designed criteria
Checking availability	confirmed	To avoid double booking
Required patient details	confirmed	To know patient needs
Easy edit booking details	Provisional	May need to edit time or history

4. Questions for the client

1. What type of questions should be asked by the patient?
2. What type of data should the practitioner be looking for
3. Should the system record patient details

4. Who should be allowed to edit an appointment?
5. Should the system be able to prevent double booking?

5. What we do not yet know

1. Is the system secured to share details?
2. Who can maintain all the data?
3. The accuracy of shared data by.

AI Activity Card - Ask, Check, Explain

Before AI

What do I think the code does? What problems can I already identify?

- The code solves the general inquiry of every patient and their needs. It solves some existing problems as well as improvises some existing features.
- The problems that I could identify are: data accuracy, system vulnerability

AI request

Act as a tutor. Explain this code and identify potential problems. Do not provide a complete replacement. Ask me questions that help me reason about the solution.

- **1. Welcome message**
- The program starts by printing a simple greeting so the user knows what the program is about.
- **2. Hard-coded appointments**
- You manually define:
 - patient name
 - practitioner name
 - appointment time
- Then print each appointment using an f-string.
- This is basic input/output — good for beginners — but not flexible.
- **3. Enhanced version using lists and dictionaries**
- You create an empty list called appointments. Each appointment is stored as a dictionary with keys:
 - "patient"
 - "practitioner"
 - "time"
- This is a much better structure than separate variables.
- **4. book_appointment function**
- This function:
 - Checks that the patient name is not empty
 - Creates a dictionary
 - Appends it to the appointments list
 - It's simple and readable.
- **5. display_appointments function**
- This function:
 - Checks if the list is empty
 - Loops through each appointment

- Prints the details
- This is a good introduction to iteration and data structures.

Evaluate

Suggestion	Useful	Unclear	Incorrect	Out of scope
Clarify design		yes		
Program must handle multiuser	yes			
Identify input and user	yes			

Decide

For each significant suggestion: Accept / Modify / Reject / Keep unverified.

Verify

- Run the code- works
- Test normal input- works
- Test unusual input- does not work
- Compare with requirements- matches
- Ask tutor/peer
- Check documentation- done

Explain

Can I explain the final code without reading the AI response? What do I still need to understand?

- Yes, I can explain the entire code without reading the AI response. Its simple and efficient.

But what I still need to understand is, does the system ensure the accuracy of the given data and does it prevent double booking by itself, does it send a reminder to the patients of their booking.